Sleeping Machines — investor pitch

The next general-purpose substrate for machine intelligence.

Sleeping Machines aims to make intelligence composable within **one trainable computing substrate**. Language and reasoning, perception and world models, embodied action, typed tables and interleaved process streams become different forms of experience available to the same learner. The ambition extends through continual learning, communication and self-design to the hardware that executes them.

An observation enters through an interface that respects its meaning: a token, a sensory event, a typed comparison or an action. It becomes an addressed message interacting with persistent memory. Learned delays and temporal races decide which evidence meets and which computation happens. Small messages carry new evidence into deeper state; separate keys and values distinguish where evidence goes from what it says. Counterfactual credit teaches the routes and writes that could have happened.

The larger promise is that these are **reusable learned computations**. Comparison, binding, retention, prediction and planning learned from one form of experience can support another through shared representations and end-to-end credit. Perception can inform reasoning; reasoning can guide action; process traces and tables can contribute to the same world model. Joint training must measure that transfer and the retention of existing skills.

This also connects the learner to its execution. Time performs computation, including useful sleeps; local events can advance it without a mandatory global clock. Inference and learning are designed around the same temporal, addressed substrate. Available memory and skill can grow beyond the activity recruited for one observation, with work tuned from small data and constrained devices to large datasets and datacenters. Candidate discovery and alternative learning work count in that budget.

A revealing proof of the platform: mixed-type tables, token sequences and anonymous event streams learned through the same temporal core, followed by measured skill transfer. The program spans small-data usefulness and large-data scaling.

Online market forecasting: confirmed on a new month

A preregistered out-of-sample win while every model keeps learning. On Binance BTCUSDT public trades, our native temporal learner, an online LSTM and two online Transformers each adapt their weights after every 32 revealed trades and are scored before any outcome can change them. All model choices were fixed from August evidence and committed before the 1 October data were downloaded; each learner carried its full August memory and optimizer state across the five-week gap.

Model (all learn online)	1 Oct loss, 65,536 trades	25 Aug loss, 65,536 trades	Online arithmetic
Sleeping Machines	**8.0785**	**5.1759**	**8.5 GFLOPs**
LSTM	8.1027	5.1967	9.5 GFLOPs
Transformer, cached history	8.1792	5.2804	124.7 GFLOPs (14.7×)
Transformer, re-encoded history	8.1764	5.2797	7,963 GFLOPs (936×)

Lower loss is better (nats per trade: next-trade timing, aggressor side and price direction). Our model leads in **all 16 of 16** consecutive 4,096-trade windows of the new month, with **14.7× less arithmetic** than the cached Transformer and **16× less retained state** (0.146 MB of weights plus prediction memory versus 2.334 MB). This is the capability a deployed forecaster needs: keep learning from a changing stream at low cost. Single seed; the sealed September test stays unopened until seed

replication. Next: return-distribution forecasts of the price itself (10 s, 1 min, 5 min), now in development. No trading-profit or measured chip-energy claim. Source-bound evidence.

Industrial fault detection: sealed win on interleaved factory logs

Predictive maintenance on anonymous machine logs. In our FAS benchmark (assembly-line simulator, two interleaved lines of 30 items, no item labels), a failing machine shows only as production steps that slowly take longer. Our model learns each step's duration from normal logs, binds every event to the earlier event it most likely continues through a race of delayed clocks, and keeps a separate running anomaly score per step. The full plan was committed before the test data were generated; each model was scored once.

Detector (2,000 normal + 2,000 faulty runs)	Fault-detection AUROC
Sleeping Machines (three independent fits)	**0.801**
Time-encoded Transformer (three seeds)	0.704
Best of nine classical detectors	0.676
LSTM (three seeds)	0.667

Ahead by 0.097 AUROC (95% interval 0.088 to 0.121), every fit above every reference (wear faults 0.772 vs 0.650; retry faults 0.829 vs 0.758). No item identities are used and the model's knowledge is fixed during scoring. Source-bound evidence.

Language: addressed memory leads

Addressed memory is our strongest language lever. On the same 8,192 FineWeb development targets (GPT-2 tokens, 1,024-token history), our keyed temporal-memory model, whose events write keyed messages that later tokens read directly, scores **5.60 and 5.62 nats/token** (two seeds) against **5.978** for the earlier flagship; removing the keyed read costs 0.10 at 1M tokens. A 6,248-parameter addressed episodic memory with read-time credit lifts the flagship itself to **5.634 (+0.345)**, while an identical plain continuation does not improve; the learner's use of history older than 64 tokens rises from 0.004–0.008 to **0.227 nats**, and it learns on its own to discount memories from earlier documents. The same principle, explicit addressed access to the right past events, produced the sealed fault-detection win above. Source-bound evidence.

Learning more efficiently. Sparse counterfactual teaching improves development NLL from **6.621 to 6.528 nats/token** while cutting complete CPU training time from **9,129 to 7,734 seconds (15.3%)** at 8.39M GPT-2 FineWeb targets, all core, discovery, backward, replay and optimizer time included. Source-bound evidence.

Forward computation teaches credit — current status and potential

Prospective real-token calibration reduces credit error **98.4% versus static credit and 97.4% versus reference-only credit**, improving all eight fresh TRAIN-tail prefixes. Shared forward-derived geometry tracks alternative parameter and Adam-moment trajectories. A separate matched four-prefix, three-repeat acquisition pipeline is **2.09x faster** than exhaustive alternative learning, with **93.5% fewer supported arithmetic operations**, including context, route discovery and exact factual/reference losses. In a fixed connected 37,377-parameter controller, conditional action-gradient noise is **77.8% lower** than static credit, with exact mean-gradient parity.

Scope: component evidence on a fixed two-layer pretrained native model, 50,257 GPT-2 token classes,64-dimensional lexical learners and65 candidate routes. Accuracy uses eight new48-target prefixes; execution and controller noise use four earlier prefixes. Source pretraining, oracle acquisition and numerical gates are additional. Supported arithmetic omits unsupported work included in wall. Early small route-policy fits have no confirmed held-out prediction gain. Completed evidence and formulas.

The opportunity is to make continual adaptation economical: models that learn as data changes, capable local machines and industrial event systems, and reusable learned credit across the general-purpose substrate. The sparse asynchronous hardware path targets memory, communication and energy budgets. These are conditional platform implications, not measured customer demand or industry-wide savings. The next gate is **better held-out prediction per total learning cost**, followed by larger data/depth and online application confirmation.

Wins and advances at a glance

Nine public leaderboard wins, each on the official splits with a sealed test scored once per run, 40 of 41 independent runs ahead of the best published result:

Benchmark	Ours	Best published	At what cost
EasyTPP Taxi (mobility events)	0.525 ± 0.001	0.522 (S2P2, NeurIPS 2025)	1/12 of the leader's parameters and compute; **reproduced independently on separate hardware**
EasyTPP Taobao (shopping)	1.399 ± 0.003	1.318 (IFTPP)	0.92× the leader's compute; largest margin (+0.081)
EasyTPP StackOverflow (Q&A activity)	−2.153 ± 0.005	−2.163 (S2P2)	**matched size and compute**; all 5 seeds ahead
EasyTPP Retweet (social cascades)	−6.326 ± 0.001	−6.348 (NHP)	**1/15** of S2P2's parameters and compute
EasyTPP Amazon (shopping reviews)	0.803 ± 0.001	0.781 (S2P2)	0.29× the leader's compute
P19 ICU sepsis prediction	AUPRC 0.639, AUROC 0.916	0.583 / 0.903 (MTM)	temporal memory adds +0.072 AUPRC on every split beyond summary statistics; reproduced
PAM wearable activity recognition	accuracy 0.978, F1 0.980	0.975 / 0.976 (MTM)	46K parameters vs MTM's 873K
TGB tgbn-trade (temporal graph: node affinity, world trade)	NDCG@10 0.868 ± 0.0005	0.863 (NAVIS, ICLR 2026)	2,107 parameters, 4 CPU-minutes; **a new domain**: interaction graphs, where heuristics beat every temporal GNN
TGB tgbl-wiki (temporal graph: link prediction, Wikipedia edits)	MRR 0.835 ± 0.0003	0.827 (TPNet)	7,995 parameters; exact per-event state; 4.5 ms per query on one CPU thread

Rigor: pre-registered reporting rules; the benchmark's own scorer agrees; a recording-grid audit every win passes; a one-command kit for third-party reproduction. **One configuration wins all five:** a single configuration with no per-dataset tuning is ahead of the best published result on all five EasyTPP datasets under a pre-registered protocol (24 of 25 seeds ahead): Taobao 1.397, Taxi 0.526, StackOverflow −2.144, Retweet −6.324, Amazon 0.802.

Advances beyond the leaderboards:

- **Tokenized language:** 6.009 ± 0.010 nats per token vs a Kneser–Ney trigram's 6.537 on GPT-2-tokenized FineWeb (3 seeds); at 4× the data 5.510 vs 6.100, so the lead grows with data.

- **Binding and recall:** associative recall across irregular gaps 97.5% (3 seeds) vs a 14% baseline, 91.6% at twice the training length; **learning through the race alone** (no backpropagation into the network) reaches 77%.

- **Learning without a backward sweep or weight transport:** a race-of-clocks event model learns from each event as it arrives, with exact memory credit carried forward (no stored history). On Taxi development data at the same 20 data passes it scores 0.4805 nats/event vs 0.4557 for backpropagation through time (3 seeds). On Taxi it matches backpropagation run to convergence (0.4774 after 100 passes) with one fifth of the data passes. Across three datasets it leads at equal passes on Taxi (+0.025) and StackOverflow (+0.046) and is level on Amazon (+0.002), while backpropagation run to 100 passes ends ahead on StackOverflow and Amazon. Separate backward

weights trained by the forward pass's own updates align to the exact dual (cosine 1.000) and learn as well as shared weights (0.4715 vs 0.4736); fixed random feedback fails (0.17). This is the learning half of clockless hardware.

- **Generative mode:** the same race-of-clocks model samples realistic event streams exactly (no rejection), for simulation, what-if analysis and synthetic data; taxi streams match real timing (gap KS 0.014 vs 0.063 naive) and event-to-event structure (0.024 vs 0.53).

- **Exact training signal and measured speed:** the closed-form likelihood removes the Monte Carlo gradient noise that sampled-intensity models carry (18% relative noise at the common 10-sample setting, measured); a streaming per-event update runs at about 1,500 events per second on one CPU core in plain PyTorch (Taxi; 20.7K multiply-adds per event vs the leader's 250K). A race of clocks provably approximates any continuous inter-event density, confirmed numerically; likelihood saturates once the number of clocks covers the data's gap modes (predicted, then confirmed), so model size follows from the data.

- **Reasoning from few examples:** temporal event chains 99.7% vs Transformers 33–41%; race retrieval 100% at 4× context.

- **Character language:** beats tuned Transformers at equal or lower compute (text8, 10M characters).

- **Learning signal and capacity:** credit to unchosen routes cuts error by 0.14 bits per character for 0.3% more work; doubling memory slots lowers error with the same eight writes per input.

- **Generality:** one shared temporal core serves four event domains; anonymous interleaved process logs (FAS v1) beat six generic detectors across three seeds.

- **Theory:** a race of clocks contains softmax attention and Mamba-style selective state spaces exactly.

New event domains:

- **Temporal Graph Benchmark, node affinity: won** (table). Pre-registered sealed test, 3 seeds: 0.8683 / 0.8674 / 0.8683, every seed above NAVIS's 0.863; persistent forecast 0.855. Validation 0.874 vs 0.860 had predicted it.

- **Temporal Graph Benchmark, link prediction: won** (table). Exact per-event state lifted our model from 0.776 to 0.852 validation MRR (theory note 159 predicted at least +0.03); three pre-registered sealed seeds score 0.8353 / 0.8350 / 0.8356 on test, every one above TPNet's 0.827, with 7,995 parameters. On tgbl-review a training-free decayed event state already reaches 0.344 validation MRR, above TGN, TGAT and TNCN (0.313–0.325); GraphMixer leads at 0.428.

- **Second neural-TPP benchmark** (Bosser & Ben Taieb; seven datasets, five fixed splits): our current event model, with repaired clocks and the addressed episodic memory that drives our language and fault-detection gains, is in development on it now; its verdicts follow when each dataset's five test splits are scored.

Where we are not yet ahead: large-scale language (our best token model reaches 5.60 nats per token after 4M training tokens on single-thread CPUs; depth scaling is under test), P12 mortality (0.871 vs 0.880 AUROC) and real-world tables.

Evidence anchors for the platform

Token sequences, anonymous interleaved process logs and mixed-type comparisons now learn within the temporal/sparse family. They exercise different information structures through addressed messages, persistent memory, temporal computation and credit to unrealized alternatives. This is evidence for a common computing construction spanning small-data tables, asynchronous analytics and tokenized language.

Evidence front	Completed indication	What it establishes
Public benchmark: timed event data (EasyTPP, ICLR 2024)	**Taobao 1.399 ± 0.003 vs best published 1.318** nats/event (+0.081) at 0.92× the leader's compute; **Taxi**: a 5-seed mixture (0.536) beats every published model at 0.41× S2P2's compute (NeurIPS 2025), and a single model leads S2P2 on the mean (0.525 vs 0.522) at 1/12 of its parameters and compute. **StackOverflow** −2.144 ± 0.004 vs −2.163 (+0.019) at 1.26× S2P2's compute. **Retweet −6.326 ± 0.001 vs best published −6.348** (NHP; S2P2 −6.365) at **1/15 of S2P2's parameters and per-event compute**. Official splits, sealed test, 5 seeds.	**All five datasets won on a public leaderboard of the family's home field, and one model configuration wins all five without per-dataset tuning**, four of them at a fraction of the state of the art's compute; the Taxi win reproduced independently.
Public benchmark: ICU sepsis prediction (P19, PhysioNet 2019)	**AUROC 0.916 ± 0.022, AUPRC 0.639 ± 0.039** vs best published 0.903 / 0.583 (MTM, 2025) on the five official splits; 62,681 parameters. Without its temporal memory the same network scores 0.900 / 0.572 on every split; trees on the same statistics 0.914 / 0.618: the temporal memory carries signal the statistics lack (AUPRC ahead of the trees on every split). Reproduced on separate hardware.	**A second public win, in clinical early warning**, with AUPRC (the clinically relevant metric at 4% prevalence) ahead beyond split spread.
Public benchmark: wearable activity recognition (PAM)	**Accuracy 0.978 ± 0.007, F1 0.980 ± 0.008** vs best published 0.975 / 0.976 (MTM, 2025), five official splits, 46,316 parameters; four of five splits ahead.	**A third public domain won**: wearable sensor streams.
One family across data types	The P19 classifier reuses the event model's temporal memory layer (same code, same size).	Wins on generative event modelling and on sparse clinical classification; the EasyTPP leader S2P2 is shown on one task type. Transfer tests are next (generality plan).
Industrial fault detection (FAS, sealed fresh draw)	**AUROC 0.801 vs Transformer 0.704**, LSTM 0.667, best classical 0.676; every native fit above every reference.	Event binding through time turns better log modelling into earlier fault detection.
Mixed-type tables, small data	**100% accuracy on 256 synthetic DEV rows from 64 FIT rows**,6,370 learned parameters. Numeric, categorical, Boolean and missingness comparisons enter before neural message processing.	The integrated core learns the mixed-type interaction; relabeling, column-order, missing-value semantics and actual alternative-write credit checks pass. Fixed predicates, one seed; learned discovery and real tables versus trees are the next tests.
Online market forecasting (preregistered new-month confirmation)	**8.0785 vs LSTM 8.1027 and Transformers 8.1764–8.1792** nats/trade on 65,536 unseen 1 October trades, every model learning online; leads all 16 of 16 windows at 14.7× less arithmetic than the cached Transformer.	Continual learning from a changing stream, cheaper than attention; choices fixed before the data existed.
Properly tokenized language	The temporal-memory token model with the keyed predecessor read scores **6.009 ± 0.010** nats per token (3 seeds) at 1M GPT-2 tokens of FineWeb vs a Kneser–Ney trigram's 6.537; at 4M tokens **5.510** vs 6.100.	First tokenized-language evidence that persistent memory binds context; the lead grows with data. A published small Transformer under the same protocol is the next reference.

The unifying advantage is the reusable construction: type-respecting comparisons and tokens become events; events recruit meaningful state and teach hard routes through their consequences. Available capacity, selected activity, temporal learning and execution are separate design axes. The larger ambition connects language and reasoning, multimodal world models, embodiment, continual learning, communication, self-design and clockless hardware through this substrate. Joint training with shared parameters must next measure skill transfer and donor-skill retention. These completed instances were trained separately.

What is already shown (language rows single-seed; native compute traced, reference compute shape-estimated):

- **Wins a public leaderboard on its home field:** EasyTPP Taobao (+0.081 nats/event over the best published model at 0.92× the leader's compute), Taxi (beats every published model at 0.41× S2P2's compute) and StackOverflow (+0.019 at 1.26× compute); 5 seeds, sealed test.

- **Wins a public clinical benchmark with the same core:** P19 sepsis prediction, AUPRC 0.639 vs 0.583 for the best published model (MTM, 2025), five official splits. The classifier reuses the event model's temporal memory layer.

- **Beats Transformers at a fraction of their compute.** On 10M characters of text8 the native model reaches 1.888 bpc against the 4-pass Transformer-256×4's 1.908, at 0.40× its training and 0.18× its inference compute, and beats the validation-selected tuned Transformers at both budgets (1.888 vs 1.996; 1.955 vs 2.215). Tuned small LSTMs lead at these budgets by 0.04–0.06 bpc.

- **Counterfactual route credit works:** 2.507 → 2.371 bpc for 0.3% extra training work.

- **Capacity beyond activity:** doubling the receiver pool improves 2.371 → 2.345 bpc at the same selected work.

- **Learned temporal computation:** 99.7–99.9% event-order accuracy from 2,000 examples seen once (Transformers 33–41%); 95.3% on timing-only discrimination where any order-only model is capped at 50%; 100% on unseen modular rules; 100% retrieval at four times the training context.

- **Scales with data and adapts online:** at 90M characters the native model reaches 1.783 bpc and beats a tuned Transformer-256×4 (1.811) at 1.06× its compute, while a tuned Transformer-192×4 (1.704) still leads; online adaptation 3.191 → 3.096 bpc.

- **Hardware economics:** in the cost model, 5.8× fewer bytes moved per character than Transformer-256×4 at better quality (modelled; silicon measurement is a funded milestone).

Typed learning: preserve meaning, then compose evidence

A category label, a temperature, a duration and a missing observation support different operations. Our typed interface applies meaningful comparisons before their outcomes enter addressed messages and persistent neural state. Computational races select evidence; counterfactual consequences teach the routes. Neural sums remain useful once representations share a declared latent space.

The proposed advantage combines tree-compatible conditional decisions with deep temporal memory and a common interface across modalities. Valid changes of units or category labels should preserve the evidence and downstream computation; training requires compatible parameter and optimizer coordinates too. The new theory derives these conditions and retains credit through computational time. **39 mathematical/interface contracts pass.** The integrated synthetic mixed-type fit already reaches 100% on 256 DEV rows from 64 FIT rows with fixed predicates. Learned predicate discovery, real tables versus trees and shared-core skill transfer are the next proofs.

Modern tabular neural models already use typed feature maps; comparison layers and neural/tree hybrids have precedents. Our contribution is the complete temporal/sparse composition and its tested consequences. See typed semantics and race composition.

Paths to revenue. Two entry routes share the core: datacenter AI at lower total cost (models and runtime), and event-native vertical models on existing CPUs and GPUs for interleaved, timestamped logs in industry, IoT, IT operations, security and finance, priced on detection value, the shortest path to revenue. Licensed runtime and accelerator IP and learning-capable event processors for mobile and robotics follow. The largest outcome is ownership of a widely used AI infrastructure layer.

The raise. €3M at €50M priced pre-money, with €100M as a stretch scenario. The 18-month program buys the team, the decisive proofs (race attention against competent Transformers at scale on GPU, the pre-registered FAS v2 home-field benchmark and a real-data track, trained sparse serving), hardware

feasibility on FPGA and neuromorphic silicon, and company-owned IP through patent review and selective filings.

Read the valuation rationale, full investment case, opportunity register and research evidence.